

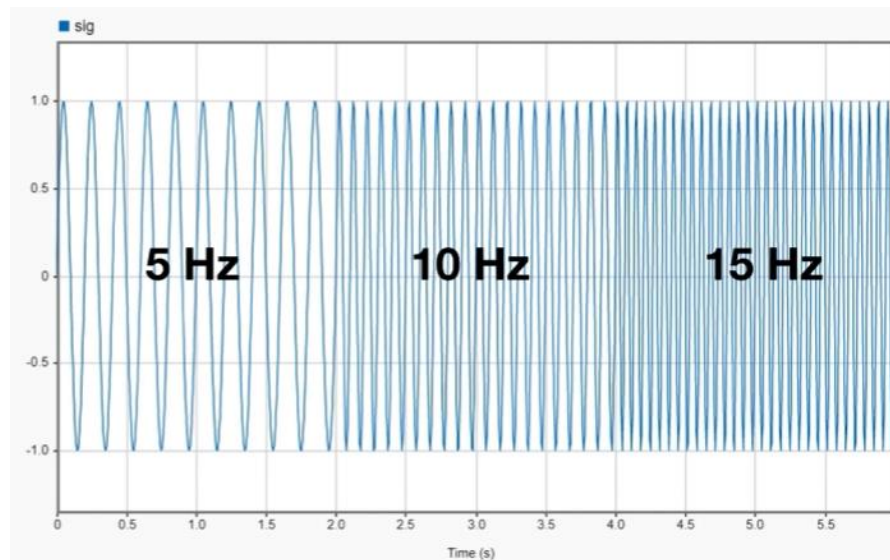
Signal Processing Using MATLAB- Assignment 1

Assignment: Implement the MATLAB code as in following script file, should be named MATLAB file as (HW1.m), Make sure to follow the following requirements:

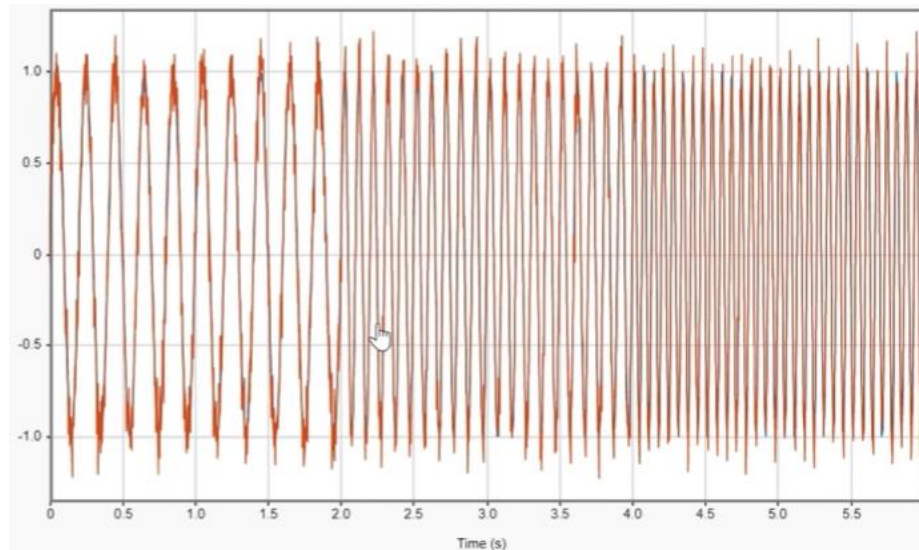
Note: Re-write the MATLAB scripts based on our course rules, make cells, cell title, and in line descripts and units as discussed in our lecture.

Part 1: Generate Signals (75 points)

- 1.1. Write a code that generates a signal (A) with three frequencies 5 Hz, 10 Hz, & 15 Hz as in the following Fig. 1 (y-label Power [mW], x-label time (s))



- 1.2. Write a code that generates a noisy signal (B) with three frequencies 5 Hz, 10 Hz, & 15 Hz using "rand function" as in the following 1 (y-label Power [mW], x-label time (s))



1.3. Save your signals A and B in one xlsx file named as signals.xlsx.

For example

```
filename = 'testdata.xlsx';  
A = [12.7 5.02 -98 63.9 0 -.2 56];  
xlswrite(filename,A)
```

Part B: Use Spectrum Analyzer app in MATLAB to analyze the signals in time and spectrum domains as follows (25 points):

